Meidie Fei

Security Analyst Candidate | Secure Software | Vulnerability Assessment | Python

Melbourne, Australia | Open to Bangkok-based 6-month contract

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SUMMARY

Early-career cybersecurity practitioner completing a Master of Cyber Security at RMIT University, with a Computer Science background and hands-on work across secure software, phishing detection, cryptography, vulnerability assessment, and cloud-backed web applications. Strong fit for security finding triage, code review support, vulnerability remediation tracking, and fast learning in a modern cloud and CI/CD environment.

CORE SKILLS

- Security: OWASP Top 10, web application security, vulnerability assessment, penetration testing labs, threat modeling, secure code review, remediation tracking
- Detection: phishing analysis, SPF/DKIM/DMARC analysis, IOC extraction, STIX 2.1 export, URL reputation, domain intelligence, sandboxing concepts
- Secure engineering: authentication, RBAC, CSRF protection, MFA, rate limiting, security headers, WAF rules, audit logging, encryption at rest, key management
- Tools: Python, JavaScript, TypeScript, SQL, Flask, FastAPI, SQLAlchemy, React, Docker, GitHub Actions, Supabase, Netlify, Cloudflare, HashiCorp Vault, Playwright, pytest, Vitest
- Communication: security documentation, test reports, technical teaching, English and Indonesian

EDUCATION

RMIT University - Master of Cyber Security

2025 - 2027 expected

Transcript highlights: Program GPA 3.3; HD in Ethical Hacking and Security Testing, Secure Software Systems, Practical Data Science with Python, and Data Communications; DI in Digital Risk Management, Security in Computing and IT, and Introduction to Cyber Security.

BINUS University - Bachelor of Computer Science

Completed

SECURITY PROJECTS

SecureVote - Secure Electronic Voting Platform

Flask, SQLAlchemy, MySQL, Docker, HashiCorp Vault, ModSecurity CRS, pytest

GitHub

- Built and enhanced a secure online voting platform using RSA blind signatures, ChaCha20-Poly1305 PII encryption, HMAC-SHA256 blind indexing, HMAC-backed audit logs, RBAC, CSRF protection, MFA, rate limiting, ModSecurity CRS, and Vault Transit result signing.
- Implemented production safety gates, environment detection, security headers, WAF integration, split database connection patterns, and GitHub Actions security test documentation.
- Verified behavior with 103 pytest tests, including blind signature protocol tests, password policy tests, PII encryption/access tests, and 10-thread voting concurrency tests.
- Added pagination security tests blocking attempts to exceed the 40-record server-side limit, including direct parameter manipulation and DoS-style per_page=99999 bypass attempts.

Automated Phishing Detection Pipeline

Python, FastAPI, asyncio, SQLAlchemy, Playwright, threat intelligence APIs

Live Demo | GitHub

- Built a 5-stage async pipeline for email ingestion, feature extraction, concurrent analysis, weighted decision scoring, and analyst feedback.
- Implemented 7 concurrent analyzer types covering header analysis, URL reputation, domain intelligence, URL detonation, brand impersonation, attachment sandboxing, and NLP intent classification.
- Supported SPF/DKIM/DMARC checks, MIME parsing, URL extraction/defanging, QR decoding, attachment classification, STIX 2.1 IOC export, JSON/HTML reporting, and dashboard workflows.
- Tested against 22 synthetic brand email samples, achieving 90% recall, 91% precision, and F1 score of 0.90, with documented false positives, weak assumptions, and remediation ideas.
- Fixed scoring and stability issues including confidence dilution from "no data" analyzer outputs, non-deterministic LLM temperature behavior, unclosed API sessions, and sender profiling re-run crashes.

CryptoToolkit - Interactive Cryptography and Attack Platform

React, TypeScript, Vite, Web Crypto API, BigInt, Vitest

Live Demo | GitHub

- Built 36 client-side cryptography modules covering cryptographic workflows, number theory, protocol composition, and real attack demonstrations.
- Implemented attacks including Bleichenbacher padding oracle, AES-CBC padding oracle, ECDSA nonce reuse, GCM nonce reuse, hash length extension, Wiener's attack, Hastad broadcast, CRT-RSA fault injection, textbook RSA malleability, and Diffie-Hellman small subgroup attacks.
- Added 95 tests using NIST and RFC test vectors, including FIPS 197 AES, NIST SP 800-38D AES-GCM, FIPS 180-4 SHA-256, RFC 4231 HMAC-SHA256, and number theory edge cases.

AES-256-GCM Secure Vault

Python, cryptography, Argon2id, pytest, Hypothesis

GitHub | PyPI

- Built a Python package and CLI implementing AES-256-GCM authenticated encryption with Argon2id key derivation, AAD binding, KDF downgrade prevention, and self-contained JSON envelope payloads.
- Wrote a full threat model and tested roundtrip correctness, non-determinism, integrity failures, malformed payload rejection, KDF boundary enforcement, legacy payload support, and property-based fuzzing.

Cloudflare and Netlify DNS GitHub Action

Node.js, TypeScript, GitHub Actions, Cloudflare API, Netlify API

GitHub

- Built an idempotent GitHub Action that attaches a Netlify custom domain and upserts a matching Cloudflare CNAME record, using scoped Cloudflare DNS tokens, dry-run mode, retry behavior, SSL provisioning checks, and CI validation.

SECURITY COURSEWORK HIGHLIGHTS

- Ethical hacking: exploited and documented SQL injection login bypass, unsafe query construction leading to unauthorized data exposure, stored XSS, cookie theft, and XSS worm behavior in controlled lab environments.
- AI security and privacy: produced a membership inference threat model for an ML hiring-screen API, including trust boundaries, attacker workflow, and defense design. Reduced attack accuracy from 94.67% to 57.56% in the defended experiment using regularization, more training data, dropout, label smoothing, and fewer epochs.
- Cryptography and security in computing: implemented AES-CBC decryption, manual RSA, AES steganography, and hybrid RSA plus AES-GCM encryption tasks.
- Cloud security: completed applied cryptography and cloud security exercises including AES, ECDSA, and security calculation work.

EXPERIENCE

MDP Studio - Co-Founder and Developer

Melbourne, Australia | 2026 - Present

- Co-founded a Melbourne web design and AI implementation studio serving small business clients.
- Built production web applications and automated lead-generation workflows using HTML/CSS/JavaScript, Python, Supabase, Netlify, Apify, and deployment automation.
- Implemented Supabase Row Level Security patterns, admin CMS workflows, HTTPS-backed Netlify deployments, security headers, and automated data pipelines for client-facing systems.
- Helped deploy 123+ production pages under mdpstudio.com.au subdomains and manage domain, DNS, email, and hosting operations.

Software Programming Instructor - KodeKiddo

Indonesia

- Taught programming fundamentals to students aged 6-18, including Android development and Unity game development.
- Debugged student code in real time and translated technical concepts for non-technical learners.

Data Assistant - BKPSDM

Indonesia

- Reviewed, verified, and corrected employee data records in a government HR system.
- Built foundational experience in data integrity, consistency checking, and structured data governance.

AVAILABILITY

Available for a Bangkok-based 6-month contract and ready to learn quickly in a security operations, product security, or application security team.